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Human Digital Biomechanical Twin-Driven Ergonomic Optimization of Bass-Guitar Support Systems: Predictive Design and Experimental Validation

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Abstract

Playing-related musculoskeletal disorders (PRMDs) are highly prevalent among bass-guitar players due to sustained asymmetrical postures, repetitive finger movements, and prolonged support of instrument weight. This study proposes a Human Digital Biomechanical Twin-driven, simulation-based approach to optimize bass-guitar support systems, integrating biomechanical modelling, motion capture, and musculoskeletal simulation. A preliminary survey among 63 Italian bass-guitar players was performed to define the experimental conditions regarding posture, instrument type, and session duration. Fifteen experienced bassists participated in laboratory trials using motion capture and postural assessment tools, including MediaPipe Pose, RULA, and AnyBody Modelling System. Baseline results highlighted significant activation of the trapezius and spinal extensor muscles (19–26% MVC), confirming high ergonomic risk. Three alternative support configurations were digitally simulated, revealing that a three-point harness system (bilateral shoulder straps plus thoracic anchoring) reduced spinal stabilizer activation by 15–25% across four anthropometric percentiles. Experimental validation confirmed enhanced comfort, reduced fatigue, and improved instrument stability, with the majority of participants preferring the ergonomic configuration. These findings demonstrate the feasibility of a simulation-based, prospective, and human-centred ergonomic design framework, offering a scalable methodology to compare and optimize adaptive instrument-support systems before physical prototyping.

Keywords: Human Digital Biomechanical Twin; playing-related musculoskeletal disorders; human-centred ergonomic design

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1. Introduction

Ergonomics has progressively evolved from task optimization toward holistic Human-Centred Design (HCD), a framework integrating safety, comfort, performance, and well-being into system development. In high-performance contexts such as professional music, human–system interaction becomes particularly demanding: musicians maintain prolonged static postures while executing repetitive, asymmetric motor patterns under cognitive and emotional stress. As a result, musculoskeletal complaints are extremely common among instrumentalists and are widely recognized as an occupational health issue in the music profession [1–5].

The term playing-related musculoskeletal disorders (PRMDs) has been introduced to describe any pain, weakness, numbness, tingling, or other musculoskeletal symptoms that interfere with a musician's ability to play their instrument at the desired level [1]. Epidemiological studies report lifetime prevalence rates of PRMDs ranging from 29% to 90% among musicians, with 25.8% to 84.4% experiencing symptoms severe enough to impair performance [6]. Among music students, prevalence rates between 43% and 63% have been reported, while professional musicians show prevalence values exceeding 80% [1,7].

In recent years, musicians have increasingly been described as "artistic athletes," due to the intensity, repetition, and precision required by musical performance [8,9]. Comparisons have also been drawn between musicians and manual workers in terms of workload and exposure to repetitive strain [10]. Despite these parallels, musicians often lack the structured ergonomic assessment, preventive training, and recovery strategies that are routinely applied in sports and industrial settings. Consequently, PRMDs remain under-recognized and insufficiently addressed in both educational and professional musical environments. Several risk factors contributing to the development of PRMDs have been identified in the literature [11] and can be broadly classified as follows.

Physiological and biological factors, such as gender and age. Gender and age influence the prevalence and consequences of musculoskeletal symptoms. Previous studies have demonstrated that women report a significantly higher prevalence of symptoms than men, with longer symptom duration [12]. Conversely, focal dystonia shows a markedly higher prevalence in males (over 90%) and in women with menstrual disorders, suggesting a possible hormonal contribution to disease susceptibility [13].

Type of instrument. Musical instruments require complex hand movements and often impose awkward postures due to their shape and weight. Some instruments are inherently more asymmetrical than others. The piano, for example, is largely symmetrical, with comparable biomechanical demands on both upper limbs; the most common disorders among pianists include muscle contracture, trigger finger, and other soft-tissue overuse injuries [14]. In contrast, string instruments such as the upright bass, violin, viola, guitar, and cello require prolonged asymmetrical postures and sustained weight support, increasing the risk of musculoskeletal pain and injury [15].

Body asymmetry. Playing instruments such as the violin or flute requires controlled movements of the arms, hands, and fingers in markedly asymmetric postures. This condition may increase the risk of musculoskeletal disorders. A study involving 47 music teachers demonstrated a higher incidence of musculoskeletal problems among musicians adopting asymmetric postures compared to those using more symmetric configurations [16].

Among string instruments, the bass-guitar represents a particularly critical case. Bass-guitar players are required to support an instrument typically weighing between 4 and 5 kg using a single shoulder strap, resulting in asymmetric load transmission to the trunk and shoulder complex. In addition, bass-guitar playing involves sustained elevation and stabilization of the fretting arm, combined with repetitive finger movements and wrist deviations, which further increase biomechanical stress on the upper limbs and spinal musculature [17]. Previous studies have shown that bass players frequently report discomfort and fatigue in the trapezius region, shoulders, and lower back, suggesting a strong association between instrument support configuration, posture, and musculoskeletal strain.

Despite the high prevalence of PRMDs among bass-guitar players, biomechanical and ergonomic investigations specifically focused on this population remain limited. Traditional ergonomic approaches are largely reactive, based on observational risk assessment and post hoc correction. The emergence of Digital Human Modelling (DHM) methodologies offers a transformative shift to simulation-based and iterative ergonomic design. A Digital Human

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Model can be defined as a virtual representation of a physical system capable of simulating its behaviour under varying conditions, and it has been proposed as a core element in Ergonomics 4.0 to enhance decision-making and proactive ergonomic design processes [18]. In this study, DHM was used to reconstruct anthropometry, posture, and external load conditions, providing the kinematic foundation for subsequent analysis. When applied to human–system interaction, a Digital Human Model does not integrate biomechanical modelling to predict physiological loads. In order to override this limitation, a Human Digital Biomechanical Twin (HDBT) was developed, extending DHM by systematically coupling anthropometric and kinematic reconstruction with musculoskeletal modelling and population scaling. In this framework, the DHM provides posture and external load conditions, while the musculoskeletal layer predicts internal muscle forces and load redistribution across multiple anthropometric percentiles, and these outputs are explicitly used to guide design iteration before physical prototyping [19]. This approach enables: human-centred technological development; data-driven design iteration; inclusive optimization across anthropometric variability; and enhanced safety and resilience in complex environments. Furthermore, HDBT systems have shown potential in musculoskeletal applications where predictive modelling informs external loading and internal biomechanical responses [20]. To date, no study has applied a HDBT framework to musical ergonomics for predictive load redistribution and design iteration.

The aim of this research is to: (i) develop a Human-Centred Digital Model framework for bass-guitar performance; (ii) develop a Human Digital Biomechanical Twin to quantify biomechanical loads associated with conventional strap configurations; (iii) digitally optimize alternative load-support configurations; and (iv) experimentally validate the most promising configuration.

2. Materials and Methods

2.1. Participatory Needs Assessment

2.1.1. Survey Construction

A preliminary survey was conducted to characterize typical bass-guitar playing habits and to support the definition of the experimental protocol. The survey aimed to collect information on playing experience, session duration, perceived fatigue, musculoskeletal complaints, and instrument characteristics. Survey development was preceded by a brainstorming session involving the authors—two of whom are expert bass-guitar players with regular performance activity—and five additional expert musicians. Expert musicians were defined as individuals with more than 10 years of playing experience or formal conservatory-level training. The final survey consisted of 12 items addressing demographic information, dominant hand, playing experience, session duration, instrument type, playing posture, perceived fatigue level, affected body regions, and the presence of musculoskeletal disorders or pathologies. The survey was conducted online with using Google® Forms and completed by 63 Italian bass-guitar players.

2.1.2. Survey Results

Among the respondents, 58 were male and 5 females, with a mean age of 40 $\pm$ 14 years. In total, 13% of respondents experienced mild diseases in their right arm, 15% in their left arm, 34% in their back, 4% in their legs, and 38% did not report any musculoskeletal disorders. Most participants (92%) were right-handed, and 97% reported playing with the left hand on the fingerboard, regardless of dominant hand. The majority (85%) had more than three years of playing experience, 10% between 1 and 3 years, and 5% had less than 1 year of experience. Regarding the duration of each music session, only 3% affirmed to participate in sessions lasting less than 30 min, 32% between 30 min and 1 h, and finally

65% in sessions lasting more than one hour. Perceived post-session fatigue was rated on a 5-point scale (1 = minimum, 5 = maximum). Overall, 12% of respondents reported a fatigue level of 1, 30% reported a level of 2, 42% reported a level of 3, and 16% reported a level of 4. No participant reported maximum fatigue. All respondents indicated discomfort in muscles located in the posterior (back) or lateral (shoulder) regions of the body, while no discomfort was reported in anterior muscle groups. The most frequently affected body regions were the hands, trapezius area, and left shoulder, while the right shoulder was rarely reported.

Regarding instrument design, two main bass-guitar body shapes were identified: regular (55%) and offset (45%). While acoustically equivalent, the offset shape (Figure 1a) features an asymmetrical body and altered centre of gravity, potentially influencing load distribution and posture. The regular bass-guitar (Figure 1b), instead, features a symmetrical and traditional shape, with balanced weight distribution along the neck axis. Participants were also asked to indicate their preferred playing position by selecting from three representative postures (Figure 2). Results showed that 68% preferred position 1, 29% position 3, and only 3% position 2. These outcomes informed the selection of experimental conditions, including instrument type, playing posture, and session duration.

Figure 1. (**a**) Offset; (**b**) regular. The dashed red line highlights the asymmetrical waist alignment of the offset design.

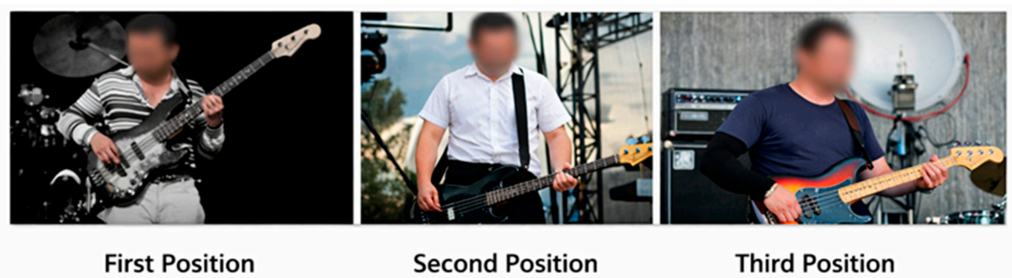


Figure 2. Bass-guitar players' common position.

2.2. Experimental Biomechanical Analysis

2.2.1. Participants

Fifteen expert bass-guitar players volunteered for the experimental study. All participants had more than eight years of experience playing the bass-guitar and regularly performed using an offset-type instrument. None reported a history of clinically diagnosed musculoskeletal disorders. All participants were male. This outcome reflects the voluntary recruitment process within the available population and does not represent an intentional exclusion criterion. Participants were informed about the experimental procedures and provided written informed consent in accordance with the ethical standards of the University of Salerno. Demographic and anthropometric characteristics were recorded to support subsequent biomechanical modelling and are reported in Table 1.

Table 1. Demographic and anthropometric data of the participants.

Characteristic	Value
Number of participants	15
Age (years)	30 ± 8.8
Body mass (kg)	76 ± 10.30
Height (cm)	173 ± 0.06
Shoulder circumference (cm)	44.36 ± 2.26
Arm length (cm)	32.72 ± 2.52
Forearm length (cm)	27.82 ± 1.40
Waist circumference (cm)	96.45 ± 7.31
Hip circumference (cm)	95.27 ± 10.45
Bass-guitar player experience (years)	9.7 ± 1.5

2.2.2. Experimental Procedure

Experiments were conducted in a controlled laboratory environment at the University of Salerno. Each participant performed a 30 min playing session using an offset-type bass-guitar. Survey results indicated that only 3% of bass players reported sessions lasting less than 30 min, 32% between 30 min and 1 h, and 65% longer than one hour. Based on these findings, a 30 min session was selected as a compromise between representing typical practice habits and ensuring participant safety. Participants played continuously, with only brief spontaneous pauses, while listening to musical tracks of their choice to promote a natural and relaxed performance. Immediately after the session, participants completed a post-test questionnaire assessing perceived fatigue on a 5-point scale (1 = minimum, 5 = maximum) for specific body regions (Figure 3), as well as the onset time and anatomical location of the first fatigue symptoms.

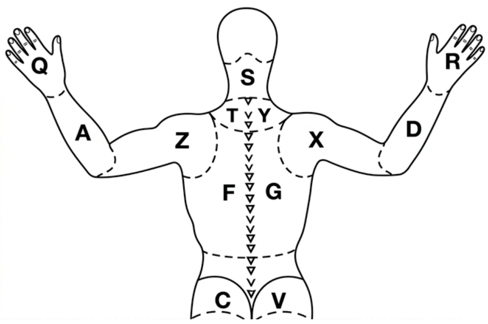


Figure 3. Questionnaire body regions. Letters denote the specific body areas used for coding participants' responses: A/D (arms), Z/X (shoulders), Q/R (hands), S (neck), T/Y (upper back), F/G (middle back), C/V (lower back).

2.2.3. Instrumentation and Software

The experimental study employed a combination of advanced hardware and software tools to capture, analyze, and simulate the biomechanical behaviour of bass-guitar players. Four high-resolution cameras (60 fps) were positioned around the participant (frontal, left, right, and overhead) to enable multi-view video recordings for motion analysis (Figure 4).

Markerless motion capture was performed using MediaPipe Pose (Google AI Edge; stable release v0.10.26), an AI-based system capable of extracting three-dimensional joint coordinates without the need for physical markers, thus enabling non-invasive and unobtrusive motion analysis (Figure 5).

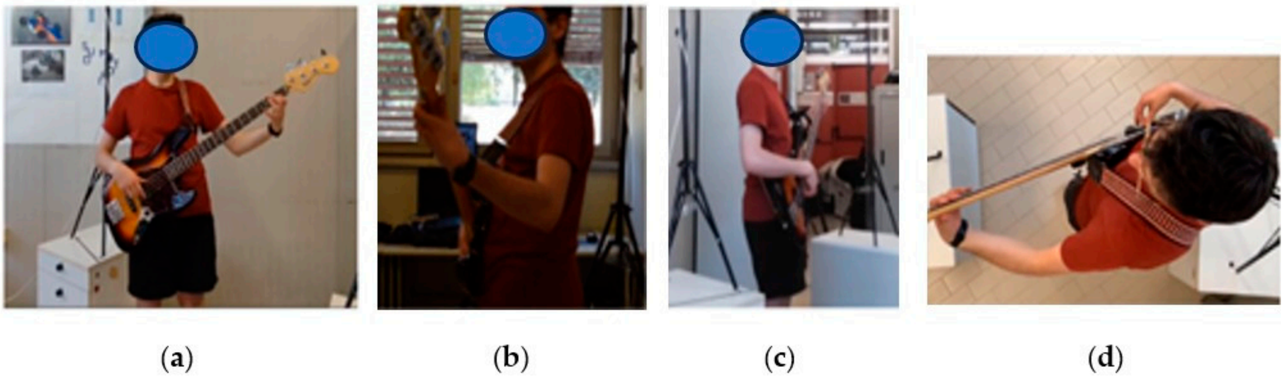


Figure 4. Video camera positions: (**a**) frontal; (**b**) left; (**c**) right; (**d**) over.

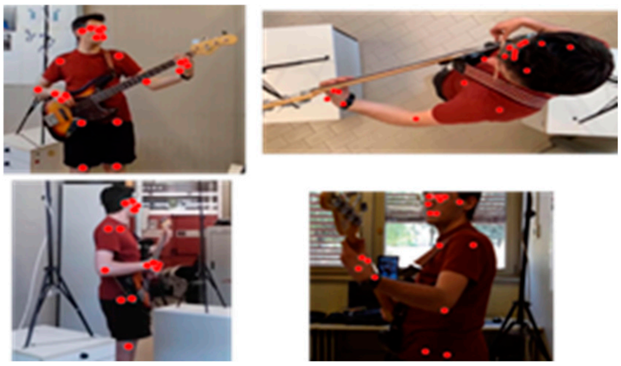


Figure 5. Photos elaboration through markers.

Postural risk assessment was conducted using the Rapid Upper Limb Assessment (RULA) method within the DELMIA V5 (version V5-6R2017, Dassault Systèmes, Vélizy-Villacoublay, France) environment, providing a structured evaluation of ergonomic stress on upper body segments. Digital Human Modelling and kinematic simulations were implemented in DELMIA, which allowed virtual reconstruction of representative postures and iterative testing of alternative instrument-support configurations (Figure 6).

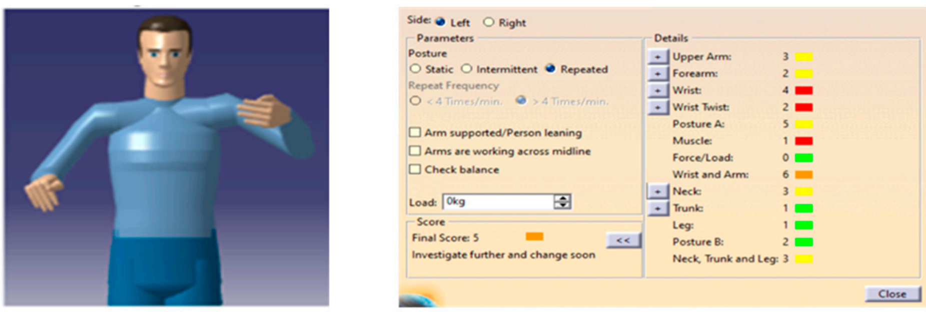


Figure 6. RULA elaboration.

Musculoskeletal load quantification was performed using the AnyBody Modelling System [21], enabling detailed simulation of muscle activation, joint reaction forces, and load redistribution across multiple anthropometric models [22].

Within this workflow, the Digital Human Model (DHM) generated in DELMIA provided the anthropometric and kinematic basis for posture reconstruction, while the Human Digital Biomechanical Twin (HDBT) extended this stage by integrating musculoskeletal modelling in AnyBody, allowing prediction of internal muscle forces and load redistribution under different support configurations.

The integrated use of these platforms allowed a comprehensive analysis of both external postural risks and internal physiological responses, supporting a predictive and Human-Centred Design approach for ergonomic optimization.

2.2.4. Experimental Results

Data analysis integrated subjective feedback, postural assessment, and musculoskeletal simulations to evaluate ergonomic risk and muscle load during bass-guitar playing.

Post-session questionnaires indicated that 63.7% of participants perceived the first symptoms of fatigue within the first 20 min of playing. The most affected regions were the left scapula (36.5%) and trapezius (27.3%), confirming the outcomes of the preliminary survey. Participants reported moderate fatigue overall, highlighting the early onset of discomfort during typical playing sessions.

Representative postures captured via MediaPipe Pose were analyzed using the RULA method within DELMIA V5. The analysis evaluated upper limb posture, trunk inclination, neck position, and external loads associated with the instrument. Across all participants, the average RULA score was 5 for both body sides, indicating a medium-to-high ergonomic risk and confirming the need for corrective intervention. Observations revealed sustained asymmetric postures, elevated fretting arms, and uneven load distribution on the shoulder and trunk. Each player was modelled in DELMIA environment with his own anthropometric characteristics.

The baseline playing posture (Configuration 0: single strap on the left shoulder) was first reconstructed and validated. External loads corresponding to the bass-guitar mass (4.5 kg) were applied at the strap contact point, while hand positions were constrained to replicate the recorded playing posture (Figure 7).

The experimental study employed a combination of advanced hardware and software tools to capture, analyze, and simulate the biomechanical behaviour of bass-guitar players.

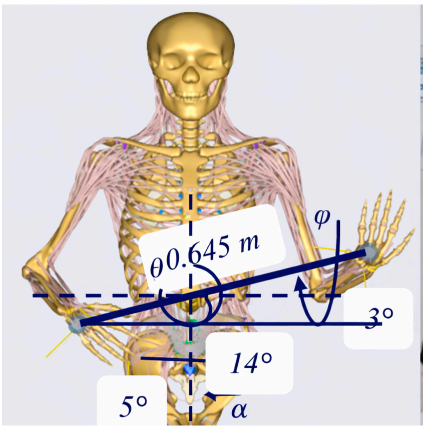


Figure 7. Configuration 0: load applied to the left scapula.

Muscle activity of the trapezius, multifidus, and erector spinae was measured in the 15 participants using the AnyBody Modelling System, with the experimental configuration reflecting typical single-strap support. Results showed considerable activation, with the trapezius reaching up to 9% of maximum voluntary contraction (MVC), and spinal extensors (multifidus and erector spinae) ranging from 18% to 26% MVC depending on the individual. These findings confirm that the bass-guitar playing posture imposes significant biomechanical loads and highlights the risk of early-onset fatigue and musculoskeletal strain.

3. Human Digital Biomechanical Twin-Driven Evaluation of Alternative Support Configurations

Building upon the baseline findings, a population-based Human Digital Biomechanical Twin was developed to enable predictive and inclusive ergonomic evaluation.

The musculoskeletal model implemented in the AnyBody® Modeling System™ (version 7.3, AnyBody Technology A/S, Aalborg, Denmark) was scaled to represent four anthropometric percentiles (5th percentile female, 50th percentile female, 50th percentile male, and 95th percentile male). This percentile-based modelling approach allowed simulation of biomechanical responses across a representative range of body sizes and segment proportions, ensuring that subsequent design solutions would be robust and inclusive rather than optimized for a single individual.

The subjective (discomfort questionnaire) and objective (postural and muscle analysis) data analysis confirmed the criticality of the bass-guitar player's task. In light of these results, to reduce the perceived fatigue during a session, it was decided to apply corrective actions directly to the instrument's support system.

Following baseline validation, three alternative support configurations were digitally implemented, iteratively simulated, and subsequently verified.

- Configuration 1: Single strap repositioned on the right shoulder (Figure 8).
- Configuration 2: Bilateral dual-strap configuration distributing load across both shoulders (Figure 9).
- Configuration 3: Three-point support system including bilateral shoulder straps and an additional anchoring point near the T5 thoracic vertebra (Figure 10).

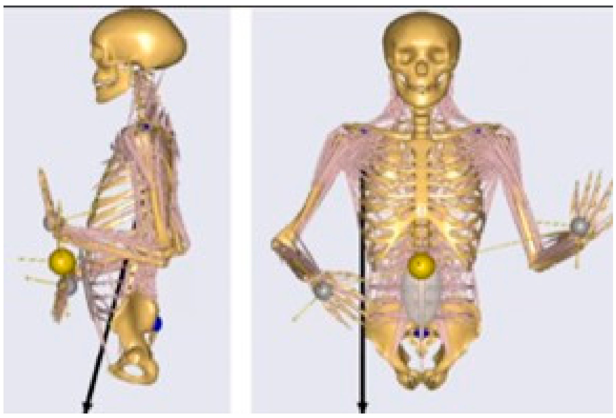


Figure 8. Configuration 1: load applied to the right scapula.

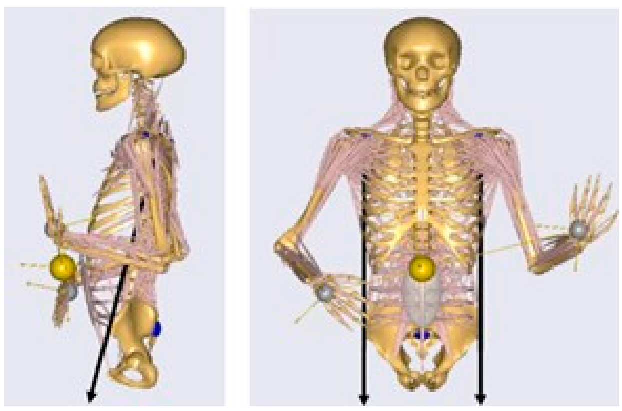


Figure 9. Configuration 2: load applied to both the left and right scapulae.

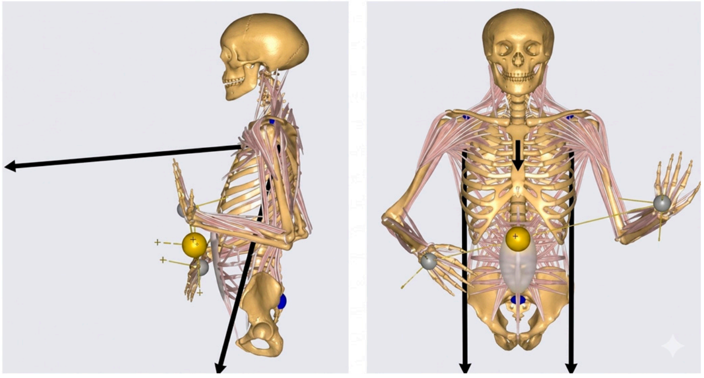


Figure 10. Configuration 3: load applied to the scapulae and near the T5 thoracic vertebra.

For each configuration, simulations were performed across the four anthropometric percentile models.

Each configuration was simulated under identical boundary conditions within the Digital Twin environment, allowing controlled comparison of internal muscle activation and load redistribution across all anthropometric models.

This structured modelling sequence—baseline validation, percentile scaling, and iterative configuration testing—positions the HDBT as an integrated simulation-based ergonomic design mediator, in which digital reconstructions of posture and internal loading are used prospectively to compare candidate support configurations before prototyping.

Muscle Load Redistribution

To quantitatively evaluate the effectiveness of the proposed support configurations, musculoskeletal simulations were conducted for all four anthropometric percentile models under identical boundary conditions.

The baseline configuration (Configuration 0) confirmed elevated activation of trapezius, multifidus, and erector spinae across all percentiles (see Table 2). Spinal extensors showed the highest activation levels, ranging from 18.7% to 25.5% MVC depending on anthropometry, while trapezius activation ranged from 6.2% to 9.0% MVC.

Table 2. Muscle activity percentage Configuration 0.

	Man 50	**Woman 50**	**Woman 5**	**Man 95**
Trapezius	6.20%	9.00%	7.30%	7.10%
Multifidus	18.50%	25.60%	22.60%	20.90%
Erector spinae	18.70%	25.50%	23.00%	20.90%

The comparative analysis of the three support configurations revealed distinct patterns of load redistribution. Global muscle activation values for each configuration are summarized in Table 3.

Table 3. Mean and standard deviation of muscle activity for each configuration and percentile models.

	Mean	**σ**	**Mean**	**σ**	**Mean**	**σ**	**Mean**	**σ**
Man 50	5.85%	5.32%	5.39%	4.99%	4.93%	4.59%	4.68%	4.34%
Woman 50	8.13%	7.29%	6.98%	6.69%	5.87%	6.27%	6.23%	6.07%
Woman 5	7.19%	6.56%	5.86%	5.92%	5.16%	4.79%	5.24%	5.15%
Man 95	6.60%	6.14%	6.34%	5.84%	5.88%	5.48%	5.38%	5.22%

Configuration 1 (single strap on the right shoulder) primarily altered the direction of load asymmetry without producing substantial changes in overall muscle demand (Figure 11).

C1 C2 C3

Figure 11. Configurations' sketches. Configuration 1 (**C1**), Configuration 2 (**C2**), Configuration 3 (**C3**).

Configuration 2 (bilateral shoulder straps) improved load sharing at the shoulder level and reduced postural asymmetry, although its effect on trunk extensor activation remained moderate (Figure 11).

Configuration 3 (three-point support system) introduced an additional thoracic anchoring point, modifying the load-transmission pathway and generating a different redistribution pattern across the musculoskeletal system. (Figure 11).

Although Configuration 2 and Configuration 3 show a significant reduction in average muscle activity compared to the traditional configuration, the overall analysis reported in Table 3 does not allow for a sufficiently sensitive distinction between the effectiveness of the two alternative solutions. To more precisely differentiate the effectiveness of the two alternative ergonomic solutions, a detailed musculoskeletal analysis was conducted on the three muscles most critically involved in maintaining posture during bass-guitar performance (multifidus, erector spinae, and trapezius). Tables 4–6 report, for each anthropometric percentile, the muscle activation values for Configurations 0, 2, and 3.

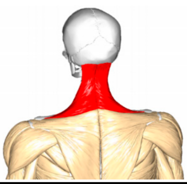
Table 4. Multifidus. Muscle activity for Configuration 0, Configuration 3 and for each percentile model.

		Configuration 0	Configuration 2	Configuration 3
	Man 50	18.50%	15.90%	15.70%
	Woman 50	25.60%	22.30%	22.20%
	Woman 5	22.60%	18.60%	18.70%
	Man 95	20.90%	19.10%	18.60%

Table 5. Erector spinae. Muscle activity for Configuration 0, Configuration 3 and for each percentile model.

		Configuration 0	Configuration 2	Configuration 3
	Man 50	18.70%	15.90%	14.90%
	Woman 50	25.50%	21.10%	21.70%
	Woman 5	23.00%	18.10%	18.10%
	Man 95	20.90%	19.30%	18.40%

Table 6. Trapezius. Muscle activity for Configuration 0, Configuration 3 and for each percentile model.

		Configuration 0	Configuration 2	Configuration 3
	Man 50	6.20%	6.10%	5.70%
	Woman 50	9.00%	7.70%	7.70%
	Woman 5	7.30%	6.40%	6.40%
	Man 95	7.10%	6.90%	6.50%

Although Configuration 2 represents a clear improvement over the traditional single-strap setup, its activation values remain consistently higher than those of Configuration 3 across all muscles and all percentile models. The differences are particularly pronounced in the spinal stabilizers (multifidus and erector spinae), where Configuration 3 achieves an additional reduction of approximately 6–15% compared to Configuration 2. Based on this quantitative evidence, across all anthropometric percentiles, Configuration 3 was selected as the optimal ergonomic solution for experimental validation.

4. Experimental Validation of the Ergonomic Configuration

The most promising configuration identified through simulation was prototyped as a wearable ergonomic harness (Figure 12).

Figure 12. Prototyped wearable ergonomic harness.

A within-subject experimental design was implemented. Each participant performed two playing sessions:

- Traditional single-strap configuration (baseline, Configuration 0).
- Ergonomic three-point support configuration (Configuration 3).

For the validation phase, the playing session was extended to 40 min to enhance sensitivity in detecting differences between ergonomic configurations under slightly prolonged conditions. Additionally, symptoms of playing-related musculoskeletal disorders typically appear during or after prolonged practice sessions, often more than 30 min [23]. The order of exposure was randomized to reduce sequence bias. Participants performed self-selected musical material to reproduce their typical playing behaviour and maintain a continuous and natural playing pattern throughout the session. Environmental conditions, instrument model, strap height, and tuning were standardized.

4.1. Participants

Fifteen bass players voluntarily participated in the experiment and an informed consent was acquired. All participants had prior experience with the instrument and regularly engaged in practice sessions involving strap use. Participation was voluntary and based on informed consent. The sample consisted predominantly of male subjects (80%), with a minority of female participants (20%). The mean age was 27.5 years (range: 18–49 years). Anthropometric parameters showed a mean height of 174.6 cm (range: 158–188 cm) and a mean body weight of 73 kg (range: 58–88 kg), resulting in a heterogeneous sample in terms of physical characteristics. In total, 60% of participants had more than 5 years of practice, 26.7% between 3 and 5 years, and 13.3% between 1 and 3 years, indicating a medium-to-high level of experience. Regarding weekly practice commitment, 46.7% reported practicing between 4 and 8 h per week, while the remaining percentages were distributed among less than 2 h (13.3%), 2–4 h (13.3%), 8–12 h (13.3%), and more than 12 h (13.3%), suggesting regular and continuous practice.

No participants reported previous muscular or skeletal disorders and 100% of participants were right-hand dominant.

Although this validation experiment involved a different group of participants compared to the preliminary phase, efforts were made to ensure comparability between the two samples. Both groups consisted of experienced bass players with regular practice habits and similar levels of expertise. Minor differences in age and anthropometric characteristics were present but fall within the typical variability observed in musician populations. The use of an independent sample was intended to minimize potential learning or expectation bias, although it introduces variability that should be considered when interpreting the results.

4.2. Subjective Evaluation and Questionnaire Data Collection

Anthropometric data (shoulder width, upper arm length, forearm length, total upper limb length, pelvic width, and thoracic circumference) were collected prior to the experimental phase.

After the trials (traditional and ergonomic strap), participants were required to compile a comparison questionnaire on perceived overall fatigue level, overall comfort, neck/shoulder muscular tension, forearm/hands tension, instrument stability, technical precision and control, naturalness and connection with the instrument and freedom of movement. Final questionnaire scores were expressed on a 5-level Likert scale structured as a direct comparison between the two conditions (1 = much worse; 2 = slightly worse; 3 = no difference; 4 = slightly better; 5 = much better).

Open-ended questions were included to collect user insights regarding: the most effective aspects of the prototype; potential areas for improvement; design suggestions for future iterations; and additional observations regarding the experimental experience. This mixed-method approach allowed the integration of quantitative subjective ratings with qualitative design feedback, supporting a human-centred evaluation of the proposed ergonomic solution.

4.3. Data Analysis and Experimental Validation Results

Data analysis was performed to evaluate perceived differences between the traditional single-strap configuration and the proposed ergonomic harness during bass-guitar playing sessions.

Statistical analysis was conducted using a one-sample Wilcoxon signed-rank test against the neutral value (3), representing no perceived difference between the traditional and ergonomic configurations. Given the ordinal nature of the Likert-scale data and the comparative structure of the questionnaire, this approach was considered appropriate. Statistical significance was set at $p < 0.05$.

The comparative questionnaire assessed perceived differences between the traditional strap and the ergonomic harness across several performance-related variables. Approximately 20% of participants reported greater comfort with the traditional strap, while 60% preferred the ergonomic harness, and 20% reported no substantial difference between the two configurations. The results indicated statistically significant improvements ($p < 0.05$) in perceived comfort, fatigue reduction, neck/shoulder tension, and instrument stability when using the ergonomic harness. Conversely, perceived freedom of movement was found to be significantly lower than the neutral value ($p < 0.05$), suggesting a potential trade-off between enhanced ergonomic support and reduced mobility.

An exploratory correlation analysis was conducted to investigate potential relationships between anthropometric characteristics and perceived ergonomic benefits.

The analysis of correlations between anthropometric characteristics (shoulder width, upper limb length, forearm length, pelvic width, and thoracic circumference) and perceived fatigue across the two configurations revealed a positive correlation between shoulder width and perceived reduction in overall perceived fatigue (r = 0.52, $p < 0.01$). Participants with broader shoulder structures reported greater improvements in fatigue reduction when using the ergonomic harness.

A weak, non-significant trend was observed between forearm length and perceived stability improvement (r = −0.56, $p > 0.05$). Given the absence of statistical significance, this observation cannot be interpreted as evidence of an anthropometric effect. Rather, it should be regarded as an exploratory indication that may warrant further investigation in future studies with larger and more heterogeneous samples.

The analysis of participants' responses regarding aspects of the proposed solution that could be improved provides useful operational indications for subsequent design iterations. The results show that the main perceived critical issues concern freedom of movement (86.7%), overall design and shape (73.3%), and ease of wearing or adjusting the device (66.7%). These findings indicate that improvements should primarily focus on the dynamic integration between the device and the performer's gestures during musical execution. A smaller proportion of participants also suggested potential improvements related to materials (40%) and overall comfort (33.3%), indicating that these aspects, although less critical, still contribute to the overall user experience.

5. Discussion

The results of this study highlight several critical ergonomic challenges faced by bass-guitar players and provide insight into how assistive support systems can mitigate musculoskeletal strain during performance. Baseline measurements revealed an early onset of fatigue within the first 20 min of playing, primarily affecting the left scapular region and trapezius muscle. These findings are consistent with previous studies indicating that asymmetric load distribution and prolonged postures generate high musculoskeletal strain among string instrument musicians [15,17].

The integration of motion capture, RULA postural assessment, and musculoskeletal simulation allowed detailed evaluation of both external postural risk and internal physiological load. Results confirmed that traditional single-strap support induces substantial trapezius and spinal extensor activation, aligning with prior research linking prolonged asymmetric postures to musculoskeletal disorders in musicians [23,24].

Within this framework, the adoption of a Human Digital Biomechanical Twin approach enabled the predictive evaluation of alternative support configurations across different anthropometric percentiles. In particular, the three-point configuration—consisting of bilateral shoulder straps combined with thoracic anchoring—consistently reduced activation levels of the trapezius, multifidus, and erector spinae muscles by approximately 15–25% MVC. These findings demonstrate how digital modelling can guide inclusive ergonomic optimization before physical prototyping. The results are consistent with recent work illustrating how HDBT frameworks extend traditional Digital Human Modelling (DHM) by integrating biomechanical modelling with predictive simulation to support inclusive ergonomic evaluation and proactive design optimization across diverse anthropometric profiles [19].

Experimental validation further confirmed the benefits of the ergonomic harness. Participants reported lower fatigue, improved comfort, and greater instrument stability during performance. Correlation analyses indicated that individual anthropometric characteristics, such as shoulder width and forearm length, modulate the perceived ergonomic benefits, emphasizing the importance of customizable support systems. These findings are consistent with emerging evidence that individualized ergonomic interventions can help prevent performance-related musculoskeletal disorders (PRMDs) and enhance the sustainability of musical performance [25], although the present results should be interpreted as preliminary due to the limited sample size. Furthermore, it should be noted that the validation phase was conducted on a different participant group than the baseline biomechanical analysis. While this choice reduced potential expectation bias, it limits the possibility of establishing a direct subject-specific correspondence between simulated predictions and experimental outcomes. Therefore, the validation results should be interpreted as providing converging evidence supporting the effectiveness of the proposed ergonomic configuration rather than as a strict one-to-one validation of the simulation outputs.

However, it should be acknowledged that the validation phase relied exclusively on subjective assessments. This choice allowed participants to play naturally without interference from sensors or markers, but it nonetheless represents a limitation. Future studies should therefore integrate objective measurements—such as electromyographic analysis or kinematic tracking—to complement subjective evaluations and strengthen the robustness of the conclusions.

At the same time, participants' feedback highlighted several aspects that require further refinement. The high percentage associated with freedom of movement suggests that the current configuration may still interfere with upper-limb kinematics or alter the performer's natural playing posture. Future design revisions should therefore focus on rationalizing the structural geometry, reducing the overall bulk of the device, and optimizing trunk contact points in order to minimize unwanted mechanical constraints. Feedback concerning the device's design and ease of adjustment further indicates the need to improve the functional ergonomics of the fastening and adaptation system. The introduction of more intuitive and modular adjustment mechanisms could enhance adaptability to different body morphologies and instrument height configurations, improving the repeatability of the optimal setup.

Additionally, suggestions related to materials and comfort indicate the potential value of adopting materials with lower local stiffness, improved breathability, and enhanced surface pressure distribution properties, as well as optimizing support interfaces to reduce localized load concentrations. From a methodological perspective, future research developments should also include objective measurements—such as electromyographic analysis, load sensors, or advanced motion capture systems—in order to more thoroughly quantify the interaction between the device and the musculoskeletal system during prolonged performance.

6. Conclusions

This study investigated the ergonomic challenges associated with bass-guitar performance and evaluated alternative instrument support configurations using a Human Digital Biomechanical Twin-driven approach. The findings confirm that bass-guitar playing exposes musicians to considerable musculoskeletal loads due to sustained asymmetric postures and the continuous support of instrument weight, factors that may contribute to early fatigue and increase the risk of playing-related musculoskeletal disorders.

By integrating HDBT methodologies with Human-Centred Design principles, the study enabled a prospective, simulation-based ergonomic design approach to evaluate different support configurations across multiple anthropometric profiles. Among the tested solutions, the three-point harness system proved to be the most effective in redistributing loads and reducing activation of key spinal stabilizing muscles, including the trapezius, multifidus, and erector spinae.

Experimental validation further indicated that the ergonomic harness improves perceived comfort, reduces fatigue, and enhances instrument stability during performance. Results should be interpreted as preliminary due to the limited and gender-imbalanced sample. Larger and more diverse cohorts will be required to confirm the generalizability of the observed effects and to strengthen the inclusivity claims of the proposed design framework. The exploratory analysis also suggested that anthropometric characteristics, such as shoulder width and forearm length, may influence the magnitude of the perceived ergonomic benefits, highlighting the potential value of adaptable support systems.

Overall, the results demonstrate the potential of digital modelling and Human Digital Twin approaches to support inclusive ergonomic design in musical contexts. Such methodologies may contribute to the development of personalized ergonomic solutions aimed at improving comfort, reducing musculoskeletal strain, and promoting more sustainable performance practices for both professional and amateur musicians.

7. Limitations and Future Research

Despite providing valuable insights into the ergonomic implications of bass-guitar support systems, the present study has several limitations that should be considered when interpreting the results.

First, the absence of female participants in the preliminary experimental phase represents a limitation of the study. This was partially mitigated through the Human Digital Biomechanical Twin approach, which included simulations across multiple anthropometric percentiles, such as the 5th and 50th percentile female models. This enabled the evaluation of biomechanical responses across a representative range of body sizes within the digital environment. However, simulated variability cannot fully replace experimental validation on a gender-balanced population; therefore, future studies should include more representative samples.

Second, the validation sample was relatively small (n = 15) and predominantly male, which limits the statistical robustness and generalizability of the findings. Future studies involving larger and more gender-balanced cohorts will be essential to fully assess how anthropometric variability influences the effectiveness of ergonomic support systems. The present results should therefore be considered preliminary evidence supporting a simulation-driven, human-centred approach to bass-guitar ergonomics.

In addition, the baseline biomechanical analysis and the validation phase were conducted on two separate groups with slightly different demographic compositions. This choice was made to avoid pre-judgement effects and expectation bias, but it also introduces a potential confounding factor, as differences between the groups may have contributed to the observed outcomes. As a result, the validation should be interpreted as providing converging, rather than strictly one-to-one, confirmation of the simulated design advantages.

Third, the evaluation relied primarily on subjective assessments collected through post-session questionnaires. While subjective perception represents an important dimension when evaluating comfort and usability, it does not directly quantify physiological or biomechanical load. Future investigations should therefore incorporate objective assessment techniques—such as electromyography (EMG), pressure distribution measurements, or extended motion capture analysis—to provide a more detailed understanding of muscular activation and load redistribution during performance. Nevertheless, EMG could represent an obtrusive measuring system while playing or freely moving, thereby affecting the global comfort and discomfort perception.

Another limitation concerns the relatively short duration of the experimental sessions. Participants interacted with the prototype for a limited time, which may not have allowed full adaptation to the new support configuration. Because musicians often develop highly specialized motor habits, longer familiarization periods may be necessary to fully evaluate the ergonomic and functional impact of new equipment. Longitudinal studies examining the use of ergonomic support systems during extended practice or rehearsal periods would therefore provide more realistic insights into their long-term effectiveness.

Finally, the prototype evaluated in this study represents an early-stage design solution. Future design iterations should focus on improving adjustability, weight distribution, and integration with natural playing movements. In this perspective, advanced Digital Human Modelling and Human Digital Biomechanical Twin frameworks could support the development of customizable or adaptive support systems optimized for individual anthropometric profiles.

Further interdisciplinary research combining ergonomics, biomechanics, product design, and digital modelling will be essential to advance the development of effective ergonomic solutions for musical performance.

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